Example 6.39 A student when walks from his house, at an average speed of 6 kmph, reaches his school ten minutes before the school starts. When his average speed is 4 kmph, he reaches his school five minutes late. If he starts to school every day at 8.00 A.M, then find (i) the distance between his house and the school (ii) the minimum average speed to reach the school on time and time taken to reach the school (iii) the time the school gate closes (iv) the pair of straight lines of his path of walk.

Solution:

Let x-axis be the time in hours and y-axis be the distance in kilometer.

From the given information, we have

$$y = 6(x - \frac{10}{60}) \;\Rightarrow\; y = 6x - 1 \tag{6.43}$$

$$y = 4(x + \frac{5}{60}) \;\Rightarrow\; y = 4x + \frac{1}{3} \tag{6.44}$$

Solving the above two equations, we get $(x, y) = \left(\frac{2}{3}, 3\right)$

$$x = \frac{2}{3} \text{ hour} = 40 \text{ minutes},\; y = 3 \text{ km}$$

(i) the distance between house and the school is 3km

(ii) the minimum average speed to reach the school on time is

$$\frac{60}{40} \times 3 = 4.5 \,\text{kmph}$$

and time taken is hours $\frac{2}{3}$ or 40 minutes

(iii) the school gate closes at 8.40 AM

(iv) the pair of straight lines of his path of walk to school is

$$(6x - y - 1)\left(4x - y + \frac{1}{3}\right) = 0$$

$$72x^2 - 30xy + 3y^2 - 6x + 2y - 1 = 0 \tag{6.45}$$

Draw the graph by using the site: https://www.geogebra.org/graphing
https://www.geogebra.org/m/fhS6HUtP

Example 6.40 If one of the straight lines of $ax^2 + 2hxy + by^2 = 0$ is perpendicular to $px + qy = 0$, then show that $ap^2 + 2hpq + bq^2 = 0$.

Solution:

Let m_1 and m_2 be the slopes of the pair of lines $ax^2 + 2hxy + by^2 = 0$ and m be the slope of $px + qy = 0$

Therefore, $m_1 + m_2 = -\frac{2h}{b}$, $m_1 m_2 = \frac{a}{b}$ and $m = -\frac{p}{q}$

since one of the straight lines of $ax^2 + 2hxy + by^2 = 0$ is perpendicular to $px + qy = 0$,